

$$\bullet \left\{ \begin{array}{l} \mathrm{d}X_t^i = \mathrm{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i = \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{array} \right.$$

Well-posedness

◦ Mean-field limit: large stochastic system $\Leftrightarrow$ deterministic PDE

◦ Lipschitz and moment-controlled ∇K and $\sigma_x \geq 0$: solutions exist up to any time and the empirical measure converges to the mean-field limit PDE

Ant stochastic analysis

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- $$\begin{cases} dX_t^i &= P v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$
- Well-posedness
 - Mean-field limit: large stochastic system $\iff$ deterministic PDE
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- Well-posedness
 - Singular (Newtonian) ∇K and $\sigma_x > 0$: $|\nabla K(x)| \sim |x|^{-1}$